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A

Project Synopsis

on

SMART BUS TRANSPORT SYSTEM (UrbanSync)

submitted as partial fulfillment for the award of

**BACHELOR OF TECHNOLOGY
DEGREE**

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Computer Science & Engineering

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Abstract

The project **Smart Bus Transport System (UrbanSync)** aims to provide a real-time, low-cost, and scalable public transport tracking solution for small and tier-2 Indian cities. Existing systems in Punjab and other states lack real-time bus visibility, leading to long waiting times, unpredictable schedules, and commuter inconvenience. UrbanSync proposes a unified digital system that integrates **real-time GPS tracking, digital ticketing, route monitoring, pass validation, and analytics dashboards.**

The solution utilizes **MQTT** for low-latency communication, **React-based UI** for conductors and commuters, and **Node.js + MongoDB/Postgres** as the backend. Existing Ticket Vending Machines (TVMs) in buses are enhanced with lightweight software to report real-time location without requiring new hardware installations. The proposed system reduces commuter waiting time by up to 30%, lowers operational costs, improves revenue transparency, and supports data-driven decision making for municipal authorities.

The approach is optimized for low-resource devices and poor network conditions, ensuring operational continuity even in remote areas. The system also contributes to environmental sustainability by increasing public transport usage and reducing private vehicle dependence.

Introduction

Public transportation in small cities often suffers from **poor visibility, undefined schedules, and inefficient route planning.** In Punjab alone, over **3000 public buses** operate daily, but a large portion lacks effective tracking and digital synchronization. Only **60%** have digital Ticket Vending Machines (TVMs) and **most lack reliable GPS tracking,** creating a significant inconvenience for passengers and authorities.

Motivation

- Commuters face unpredictability and long bus waiting times.
- Transport authorities experience leakages in revenue due to manual ticketing.
- Real-time monitoring is missing in tier-2 and tier-3 cities due to high hardware costs.

Gaps in Existing Solutions

- Existing GPS installations are **expensive** (₹5k–₹12k per bus) and not scalable.
- Smartphone-based conductor apps are prone to damage/theft, require training, and create data security issues.

- Camera-based systems are costly, complex, and raise privacy concerns.

Our Contribution – UrbanSync

UrbanSync introduces a **software-driven GPS integration** for existing TVMs, reducing hardware cost to almost zero. Key features:

- Real-time bus tracking using MQTT.
- Digital ticketing + revenue logs.
- Pass validation & fare calculation.
- Offline-first technology with IndexedDB.
- Central monitoring dashboard with analytics.

Technology Used

- **Frontend:** React, Leaflet/Map libraries
- **Backend:** Node.js, Express
- **Database:** MongoDB/Postgres
- **Real-time Communication:** MQTT Broker
- **Storage:** IndexedDB for offline mode

Field of the Project

Transportation engineering, real-time systems, IoT-enabled public mobility, smart city initiatives.

Literature Survey

1. **"Govt to launch tracking system in 1,800 buses, Punjab" – The Tribune, Business Standard**
Discusses Punjab government's plan to spend ₹5.8 cr on GPS installation.
Relevance: UrbanSync reduces this cost drastically by offering **software-only tracking**, expanding coverage without additional hardware.
2. **"Vehicle tracking system installed in PRTC buses" – The Tribune, Feb 27, 2022**
GPS systems were partially implemented in PRTC/PUNBUS, but delayed due to technical and funding issues.

Relevance: Highlights the need for low-cost, reliable alternatives like UrbanSync.

3. Alternative GPS + Camera Solutions

- Require high-cost equipment.
- Complex data processing and privacy issues.

Limitation: Not feasible for smaller cities with budget constraints.

4. Smartphone Tracking Solutions

- Expensive for conductors, frequent charging issues, security risks.
- High maintenance.

Limitation: Operationally weak.

UrbanSync overcomes these limitations with a low-cost, software-based approach integrated directly into existing ticket machines.

Methodology/ Planning of work

1. Requirement Analysis

Study existing bus operations, TVM capabilities, and real-time tracking needs.

2. System Architecture Design

- React-based UI
- Node.js backend
- MQTT-based real-time communication
- Database modeling

3. Module Development

- Real-time GPS integration
- Ticketing system
- Pass validation
- Dashboard and analytics

- Offline-first support

4. Data Flow Setup

- /bus/{BusID}/location MQTT channels
- Event logs for ticketing and conductor activity

5. Implementation & Testing

- Unit testing
- Integration testing
- Real-time performance testing

6. Deployment

- Cloud deployment
- Admin access management

Facilities required for proposed work

Hardware Requirements

- Existing TVM devices
- GPS-enabled handheld devices (if needed)
- Server hosting (cloud/VPS)

Software Requirements

- Node.js
- React.js
- MQTT Broker (Mosquitto/EMQX)
- MongoDB/Postgres
- IndexedDB (browser-based)

- Analytics Dashboard tools

References

- [1] The Tribune, "Govt to launch tracking system in 1,000 buses, Punjab," 2022.
- [2] The Tribune, "Vehicle tracking system installed in PRTC buses," Feb. 27, 2022.
- [3] Business Standard, "Punjab to implement GPS-enabled tracking system," 2022.
- [4] MQTT Documentation. www.mqtt.org
- [5] React Documentation. <https://react.dev>
- [6] Node.js Documentation. <https://nodejs.org>

SPECIFICATIONS FOR SYNOPSIS

1. The synopsis shall be computer typed (English- British, Font -Times Roman, Size-12 point) and printed on A4 size paper.
2. The Synopsis shall be typed on one side only with 1.5 space with a margin 3.5 cm on the left, 2.5 cm on the top, and 1.25 cm on the right and at bottom.
3. In the synopsis, the title page should be given first. This should be followed by index, notations/nomenclature.

4. The diagrams should be printed on a light/white background, Tabular matter should be clearly arranged.
5. The caption for Figure must be given at the BOTTOM of the Fig.
6. The caption for the Table must be given at the TOP of the Table.
7. The Bibliography/References should be written in IEEE format.


4/12/25